

Activity 4: Two virtual environments

Modern DevOps for Systems Biologists

Jun Allard

Last year you measured the brightest pixel in a microscope image, added the background offset your protocol calls for, and wrote the number in your paper. Today you run the same script on the same image, and get a different number.

A. Two answers from one script

- i. Check that you have `uv`, the tool that builds and runs Python environments:

```
uv --version
```

If that fails, install it and open a new terminal:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

- ii. Write this into a file called `brightest.py`:

```
import numpy as np

# An 8-bit microscope image. Every pixel is a whole number from 0 to 255.
image = np.array([[12, 250], [180, 7]], dtype=np.uint8)

peak = image.max()      # the brightest pixel
corrected = peak + 100  # the protocol says to add a background offset of 100

print("numpy", np.__version__)
print("brightest pixel:", peak)
print("after offset:  ", corrected)
```

- iii. Run it twice, against two different versions of numpy. `uv run --with` builds a throwaway environment containing exactly the version you name, runs your script inside it, and throws it away again:

```
uv run --with 'numpy==1.26.4' brightest.py
uv run --with 'numpy==2.3.2' brightest.py
```

iv. Write down both answers.

(If `uv` pauses to download a Python as well as a `numpy`, that is expected: the interpreter is part of the environment too, and `uv` picks one each version of `numpy` can actually run on.)

B. Which number is wrong, and why

- i. The brightest pixel is 250 in both runs, so the image is not the problem. $250 + 100 = 350$, and 350 does not fit in the 0-255 range that `uint8` can hold. One version widens the number to make it fit; the other lets it wrap around past 255 and start again from 0. Say which version did which, and check that the wrapped answer is the one you would predict.
- ii. Only one of the two runs said anything about this at all. Look again at the output of each — including anything printed before the numbers — and say what warning you got, from which version, and whether you would have noticed it in a script that printed a hundred lines.
- iii. This change is documented in [NEP 50](#), which shipped in `numpy` 2.0.

C. Say which version you meant

Make the version a property of the project.

- i. Create a project and pin the version you decided to believe:

```
uv init brightness
cd brightness
uv add 'numpy==1.26.4'      # or 2.3.2 --- your choice, but say which
```

- ii. Move `brightest.py` into that directory and run it with no `--with` at all:

```
uv run brightest.py
```

Confirm you get the answer you chose.

- iii. Two files now record the environment.

```
cat pyproject.toml
cat uv.lock
```

`pyproject.toml` is what you asked for; `uv.lock` is what you got, down to the exact build of every package numpy itself depends on. The second is the one that makes someone else's run identical to yours. `uv init` wrote a third, `.python-version`, for the same reason.

- iv. Commit both, together with the script. A collaborator who clones this and types `uv run brightest.py` now gets your number, on their machine, without asking you anything.
- v. Finally, write a `QUICKSTART.md` in that directory: four lines that take a stranger from a fresh clone to the number your script prints. Assume they have nothing installed but `git`.